

8. (5.0score)

某区域静磁场的强度 $\vec{H} = yz\vec{e}_x + 2xz\vec{e}_y + 3xy\vec{e}_z$ 则该区域的自由电流密度为

A

$$\vec{J} = x\vec{e}_x - y\vec{e}_y + z\vec{e}_z$$

B

$$\vec{J} = x\vec{e}_x + y\vec{e}_y + z\vec{e}_z$$

C

$$\vec{J} = x\vec{e}_x + y\vec{e}_y - 2z\vec{e}_z$$

D

$$\vec{J} = x\vec{e}_x - 2y\vec{e}_y + z\vec{e}_z$$

$$\nabla \times \vec{H} = \vec{J}$$

$$\begin{vmatrix} \vec{e}_x & \vec{e}_y & \vec{e}_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ yz & 2xz & 3xy \end{vmatrix}$$

$$= x\vec{e}_x - 2y\vec{e}_y + z\vec{e}_z$$

10. (5.0score)

对于一个矢势 $\vec{A}$ ，无论在什么规范下，下面边界关系总是成立的是：

$$\vec{B}$$

$$B_{1n} = B_{2n}$$

$$H_{1t} = H_{2t}$$

A

$$A_{1n} = A_{2n}$$

B

$$A_{1t} = A_{2t}$$

C

$$\vec{n} \times (\nabla \times \vec{A}_2 - \nabla \times \vec{A}_1) = \vec{a}$$

D

$$\vec{A}_1 = \vec{A}_2$$

$$\vec{B} = \nabla \times \vec{A}$$

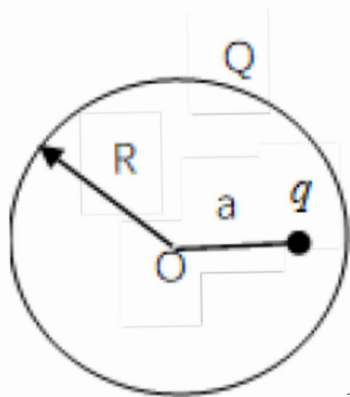
试证明在导体和介质的交界面上，电力线垂直于介质表面。

$$\oint \vec{E} \cdot d\vec{l} = 0$$

$$0 + E_t = 0 \Rightarrow \text{介质处 } E_t = 0$$

∴ 电力线垂直于介质表面。

(30分) 如图所示，在半径为R的导体球壳表面上均匀带有电荷Q，在导体球壳内距球心a的地方有一点电荷q，求空间各处的电势，球壳内表面的电荷密度分布。



内表面的电势分布为 -q.

$$r > R \text{ 时 } \oint \vec{E} \cdot d\vec{S} = \frac{Q+q}{\epsilon_0}$$

$$E = \frac{Q+q}{4\pi\epsilon_0 r^2}$$

$$\therefore U = \int_r^\infty \frac{Q+q}{4\pi\epsilon_0 r^2} dr = \frac{Q+q}{4\pi\epsilon_0 r}$$

$r < R$ 时

由镜像法的等效电荷可知

$$q' = -\frac{R}{a}q \quad q'' = \frac{R}{a}q$$

$$\therefore U = \frac{Q}{4\pi\epsilon_0 R} + \frac{q}{4\pi\epsilon_0 \sqrt{a^2 + r^2 - 2ar \cos \theta}} + \frac{-\frac{R}{a}q}{4\pi\epsilon_0 \sqrt{\frac{R^2}{a^2} + r^2 - 2r \frac{R}{a} \cos \theta}}$$

9. (5.0score)

对于一个矢量 $\vec{K}$ 和一个标量 f，考虑沿任意回路的积分，则：

A

$$\oint_L \vec{K} \cdot d\vec{l} = \oint_L (\vec{K} + \nabla f) \cdot d\vec{l}$$

$$\text{B } \oint_L \vec{K} \cdot d\vec{l} \neq \oint_L (\vec{K} + \nabla f) \cdot d\vec{l}, \text{ 但不能判断大小}$$

$$\text{C } \oint_L \vec{K} \cdot d\vec{l} > \oint_L (\vec{K} + \nabla f) \cdot d\vec{l}$$

$$\text{D } \oint_L \vec{K} \cdot d\vec{l} < \oint_L (\vec{K} + \nabla f) \cdot d\vec{l}$$

$$\nabla \times \nabla \varphi = 0$$

可知标量的度元恒。

C

W =

$$\int \rho \varphi dV$$

A

$$\frac{1}{4\pi\epsilon} \int dV \int \frac{\rho(\vec{x})\rho(\vec{x}')}{r} dV'$$

B

$$\frac{1}{2\pi\epsilon} \int dV \int \frac{\rho(\vec{x})\rho(\vec{x}')}{r} dV'$$

$$\frac{1}{2} \int \rho(\vec{x}) \cdot \int \frac{\rho}{4\pi\epsilon} dV$$

C

$$\frac{1}{8\pi\epsilon} \int dV \int \frac{\rho(\vec{x})\rho(\vec{x}')}{r} dV' \checkmark$$

D

以上都不正确

真空中的带电导体产生的电势为 φ ，则导体表面所带电荷面密度 σ 为：

A

$$-\epsilon \frac{\partial \varphi}{\partial n} \Big|_S$$

C

$$\sigma = \epsilon_0 \frac{\partial \varphi}{\partial n} \Big|_S$$

B

不能确定

C

$$-\epsilon_0 \frac{\partial \varphi}{\partial n} \Big|_S \checkmark$$

D

常数

下来标量函数中描述无电荷区域静电势的是

B $\nabla \cdot (\nabla \cdot \varphi) = 0$

A $7x^2 + 3z^2$

B $2x^2 + 3y^2 - 5z^2$ ✓

C $5x^2 + 8y^2 - 5z^3$ ✗

D $3x^3 + 6y^2$

若界面法线方向由介质 2 指向介质 1, 则静电场电势在分界面附近满足

A
$$\varepsilon_2 \frac{\partial \varphi_2}{\partial n} - \varepsilon_1 \frac{\partial \varphi_1}{\partial n} = -\sigma_f$$

B
$$\varepsilon_0 \left(\frac{\partial \varphi_2}{\partial n} - \frac{\partial \varphi_1}{\partial n} \right) = -\sigma_f$$

C
$$\varepsilon_0 \left(\frac{\partial \varphi_2}{\partial n} - \frac{\partial \varphi_1}{\partial n} \right) = \sigma_f$$

D
$$\varepsilon_2 \frac{\partial \varphi_2}{\partial n} - \varepsilon_1 \frac{\partial \varphi_1}{\partial n} = \sigma_f$$
 ✓

$$\sigma_f = -\varepsilon_2 \frac{\partial \varphi_2}{\partial n} + \varepsilon_1 \frac{\partial \varphi_1}{\partial n}$$

$$\frac{\partial \varphi}{\partial n} = -E$$

$$n_{12} \cdot (\vec{P}_1 - \vec{P}_2) \cdot \vec{n} = \sigma_f$$

3. (5.0score)

已知极化强度 $\vec{P} = xz^2\vec{e}_x + 2yz\vec{e}_y + z\vec{e}_z$, 则极化电荷密度 ρ_P 为:

A

$$\rho_P = (z+1)^2$$

B

$$\rho_P = -(z-1)^2$$

C

$$\rho_P = -(z+1)^2$$

D

$$\rho_P = (z-1)^2$$

$$\begin{aligned} -\nabla \cdot \vec{P} &= -(z^2 + 2z + 1) \\ &= -(z+1)^2 \end{aligned}$$

2. (5.0score)

对任意介质, 下方方程一定正确的是

A

$$\text{极化强度矢量 } \vec{P} = \vec{D} - \epsilon_0 \vec{E}$$

B

$$\text{极化强度矢量 } \vec{P} = (\epsilon - \epsilon_0) \vec{E}$$

C

$$\text{磁极化强度矢量 } \vec{M} = \frac{1}{\mu_0} (\mu - \mu_0) \vec{H}$$

D

$$\text{磁极化强度矢量 } \vec{M} = \chi_m \vec{H}$$

玄机在于矢量

1. (5.0score) 变化的磁场激发的感应电场满足

A

$$\nabla \cdot \vec{E} = 0, \nabla \times \vec{E} = 0$$

B

$$\nabla \cdot \vec{E} = \frac{\rho}{\epsilon_0}, \nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

C

$$\nabla \cdot \vec{E} = 0, \nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

D

$$\nabla \cdot \vec{E} = \frac{\rho}{\epsilon_0}, \nabla \times \vec{E} = 0$$